

Major Project Report

on

“CSE Stock Hub”

”



Presented By :-

Deepan Sharma (202200478)

Diya Patidar (202200059)

Navraj Ghatani (202200517)

Tejas (202200486)

Under the guidance of :-

Mr. Tawal KR Koirala

(Assistant Professor - I , CSE Department, SMIT)



SMIT SIKKIM
MANIPAL
UNIVERSITY
SIKKIM MANIPAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SIKKIM MANIPAL INSTITUTE OF TECHNOLOGY
(A constituent college of Sikkim Manipal University)
MAJITAR, RANGPO, EAST SIKKIM – 737136

Content

Sl. No	Topics	Page No
1	Abstract	3
2	Introduction	4
3	Literature Survey	5-6
4	Problem Definition	7
5	Solution Strategy	8-9
6	Methodology	10-14
7	Outcome	15-17
8	Remaining Work	18
9	Gantt Chart	19
10	References	20

ABSTRACT

Our project is about developing a web-based asset management system named **CSE StockHub** for the Computer Science and Engineering (CSE) Department of Sikkim Manipal Institute of Technology. The department maintains a large number of assets such as computers, laptops, and electrical equipment, which until now have been tracked manually using registers and spreadsheets. This traditional approach often results in duplication of entries, difficulty in updating information, and errors in record-keeping. To overcome these challenges, the CSE StockHub has been designed as a centralized and digital solution to systematically manage departmental stock.

The system provides functionalities to add new items, update or edit existing details, delete outdated records, and access a complete inventory of assets in an organized format. Each item can be stored with attributes such as category, condition, and location, enabling accurate monitoring and better utilization of resources. The web application also includes a user-friendly interface, ensuring that administrators and staff can easily navigate the system and access real-time information about departmental assets. By digitizing the management process, our project reduces manual workload, enhances transparency, and minimizes the chances of data inconsistency. Overall, the CSE StockHub contributes to improving accountability, efficiency, and long-term planning for the effective utilization of resources within the CSE Department.

INTRODUCTION

The CSE department manages a diverse range of physical and digital assets—including classroom furniture, laboratory equipment, computing devices, and shared resources—used daily by students, faculty members, and administrative staff. Currently, most inventory updates and stock verification processes are carried out manually through Excel sheets, handwritten registers, or physical inspection notes. As a result, the information becomes scattered across multiple files, inconsistently updated, and difficult to retrieve when required.

This manual workflow introduces several challenges: frequent data entry errors, delays in updating stock movement, duplicate or missing entries, and noticeable mismatches between ledger quantities and the actual ground stock. These inconsistencies make it difficult to maintain a clear audit trail and complicate the tracking of items across different locations such as labs, classrooms, faculty rooms, and the store. Recognizing these limitations, the *CSE Stock Hub* has been proposed as a unified digital platform to centralize all inventory-related information while ensuring transparency, accuracy, and real-time accessibility for authorized users.

The project employs a structured **MySQL database** to store all inventory records, a **Python backend** to process CRUD operations and authentication, and a modern **HTML/CSS/JavaScript frontend** to deliver an intuitive, user-friendly interface. This integrated system supports secure login, role-based access control, item addition and editing, deletion, discrepancy reporting, and analytical dashboards. Advanced features such as CSV-based reconciliation, faculty assignment tracking, and graphical insights further enhance asset monitoring and decision-making capabilities.

Additionally, the system incorporates automated data-cleaning routines, discrepancy detection algorithms, and audit-ready report generation. These components help streamline departmental audits and ground verification cycles, reducing workload, minimizing human error, and ensuring that the entire inventory management process becomes faster, more reliable, and easier to maintain over time.

LITERATURE SURVEY

SL.NO	Author	Paper Details and Publication details	Findings	Relevance to the project
1	Chan Chin Wei, Sathiapriya A/P Ramiah, and Nurul Farhaini Razali	Inventory Management Systems (IMS), published in the JATI Journal of Applied Technology and Innovation, Vol. 7, No. 3, (2023).	An IMS web application was developed for SMEs in Malaysia. It proved effective and efficient by automating tasks, reducing manual effort, and improving stock control. Key features include automated reports and low-stock reminders. The system was built using the Rapid Application Development (RAD) methodology and employed PHP, HTML, CSS, and JavaScript.	shows how a web-based system can provide systematic management and automation, directly addressing the issues of manual inventory. Its features are directly applicable to the system
2	Hongqi Wang.	Design of Fixed Assets Management System in Colleges and Universities Based on Barcode and PFDI Technology, published in the Journal of Physics: Conference Series, Vol. 1992 (2021).	This paper introduces a system using barcode and PFDI technology to improve fixed asset management. The system assigns a unique number to each asset for fast and accurate tracking from purchase to disposal. It also handles data management, querying, and reporting.	This paper's focus on barcode technology is directly relevant to the tracking aspect of CSE StockHub project. The use of a unique identifier for each asset provides a clear and effective method for systematic management and record-keeping, which can be a fundamental part of our physical tracking system.

3	R Setiawan, D Kurniadi, H Aulawi, and R Kurniawati	Asset management information system for higher education, published in the Journal of Physics: Conference Series, Vol. 1402 (2019).	The study's goal was to design a web-based Asset Management Information System (AMIS) prototype for higher education. The system's primary function is to manage asset data and calculate asset depreciation. The researchers used the Scrum methodology for system development to allow for changes during the process. The system includes various modules for managing asset data, locations, providers, and users, as well as generating reports. It specifically incorporates a depreciation calculation method based on the straight-line formula.	This paper is highly relevant as it provides a comprehensive model for an asset management information system. The findings on integrating a depreciation calculation are critical for the record-keeping and financial tracking components of the CSE StockHub project. The use of the Scrum methodology also offers a valuable insight into an effective approach for developing our system, ensuring it can accommodate changes and meet user expectations
---	--	---	--	---

PROBLEM DEFINITION

The Department of Computer Science and Engineering (CSE) in Sikkim Manipal Institute of Technology owns and manages a large number of assets such as computers, projectors, laboratory instruments, classroom furniture, and other equipment. These resources are essential for teaching, research, and administrative work. However, the way these assets are currently managed is mostly **manual**, using registers or spreadsheets.

This manual system of management creates several problems:

1. **Inaccurate Records** – When data is entered by hand, mistakes are common. For example, one item may be entered twice or sometimes not entered at all.
2. **Duplication of Entries** – Since there is no central control, the same asset can be listed multiple times in different registers or files.
3. **Difficulty in Tracking Usage** – If a computer or piece of lab equipment is being used, it is not always clear who is using it, for how long, or whether it is available.
4. **No Real-Time Updates** – The current system cannot immediately show whether an asset is working, under maintenance, or out of order.
5. **Lack of Accountability** – Without a centralized system, it is difficult to know who is responsible for maintaining or updating asset information.
6. **Time-Consuming Maintenance** – When equipment breaks down, tracking its repair history or scheduling maintenance becomes slow and complicated.

Because of these issues, managing the department's assets becomes inefficient, time-consuming, and costly. Faculty and staff waste valuable time searching for information about resources, while students may face delays in using essential equipment.

To solve this, there is a clear need for a **centralized digital inventory management system** that can:

- Store all asset details in one structured database.
- Track the condition and availability of each asset in real time.
- Provide quick search and filter options for easy access to information.

SOLUTION STRATEGY

The primary objective of the Inventory Management System is to create a unified and efficient platform that can maintain, track, and analyze the stock and assets present in the CSE Department. The solution strategy has been carefully designed by dividing the development into three major components: Backend System, Frontend Interface, and System Integration. Each component focuses on enhancing data accuracy, operational efficiency, and overall user experience. The complete workflow ensures that both administrative and academic needs of the department are addressed with scalability for future enhancements.

Backend Development

A well-structured backend system forms the core of the project. A **centralized MySQL database** has been developed to store all categories of inventory—including equipment, furniture, assignments, and stock updates—from various laboratories under the CSE department. The database design focuses on integrity, storage optimization, and maintainability.

During implementation, a thorough **data cleaning phase** was executed to resolve inconsistencies in existing records. This included standardizing column names, defining proper primary keys and relationships, removing duplicate entries, and validating mismatched records. These steps ensure that the stored data is reliable and supports accurate analytics.

To support decision-making and ongoing auditing, several **SQL analytical views** were created:

- **vw_furniture_status**: Summarizes furniture availability, issued items, and missing stock.
- **vw_equipment_status**: Highlights discrepancies, mismatch counts, damaged items, and calculates total assets with automated KPI generation.

For system operations, a **Python-based CRUD backend** was developed. This module enables:

- Addition of new inventory items into the system
- Updating item details such as status, lab assignment, and condition
- Removal of obsolete or non-functional items
- **CSV Export** feature for offline processing, reporting, or audits
- Dashboard **KPI reporting**, helping management track stock efficiency and department requirements in real-time

Security and controlled access are ensured through an **Authentication and Authorization system**. Three primary roles are defined:

- **Admin** – Complete access including CRUD, user management, and reconciliation
- **Lab In-charge** – Limited access to update and view assigned lab inventory
- **Viewer** – Read-only access to reports and stock information

This role-based model ensures data privacy and departmental responsibility while minimizing unauthorized changes.

Frontend Development

A professional and user-friendly web interface was developed using HTML, CSS, and JavaScript. The goal is to make the system easily accessible to users with different technical backgrounds, while offering powerful functionality for staff involvement.

The interface is fully responsive, meaning it adjusts seamlessly across desktops, laptops, and tablets. To improve visualization and clarity, Chart.js is used to render graphical representations of stock analytics and KPIs.

The major interface modules include:

- **Dashboard:** Displays real-time summary cards with inventory counts, analytics, and charts for quick insights.
- **Inventory Management:** Advanced search, filtering, pagination, and interactive pop-up modals for adding or updating entries.
- **Faculty Assignment Module:** Helps track equipment issued to faculty or individual labs, reducing misplacement of items.
- **Reconciliation Panel:** Allows CSV file upload for cross-checking physical verification reports against existing database records.
- **Reports & Analytics:** Organized tables and downloadable reports for auditing and administrative purposes.
- **Users & Settings:** Manage user roles, authentication status, and system preferences.
- **Login System:** Provides secure access with form validation and role-based redirection.

Overall, the UI is built using clean layouts, smooth animations, and intuitive navigation to deliver a modern and efficient user experience.

Integration Plan

The final phase of the strategy focuses on connecting the developed components into one fully functional application. A REST API will be implemented using Flask or FastAPI, allowing both frontend and backend to communicate securely.

Integration will enable:

- Dynamic CRUD operations directly connected to the MySQL database
- Role-based access control and secure login sessions
- Backend-powered reconciliation to highlight mismatches and update stock status
- Automatic logging of user activity for transparency and audit trails
- Removal of placeholder local storage data currently used in UI prototypes

With this integration, the system will operate in real-time, eliminating manual updates and establishing a continuous data flow between departments and administrative staff.

METHODOLOGY

The development of the CSE Stock Hub followed a systematic and structured methodology to ensure accuracy, scalability, and usability. The entire process was divided into multiple phases, starting from raw data collection and database creation to backend development, frontend integration, and system testing. Each phase played a crucial role in transforming a manual, error-prone inventory process into a fully digital, automated, and user-friendly system.

Step 1 — Data Collection and Requirement Analysis

The first phase involved collecting inventory data and understanding the functional requirements of the CSE department. Multiple stakeholders—including faculty members, lab assistants, and administrative staff—provided insights into the challenges faced during traditional inventory management.

Data Sources Included:

- Excel sheets containing equipment lists, furniture records, and lab-wise inventories
- Physical registers noting ground quantities during audits
- Faculty-specific allocation lists
- Handwritten logbooks maintained during stock movement

Key Observations:

- Data was scattered across different files and formats.
- Different versions of the same file existed among users.
- No relationship existed between datasets such as equipment, faculty assignment, and location.
- Ledger stock rarely matched the ground stock collected during audits.

From this analysis, the need for a centralized system with structured tables, standard naming conventions, relationships, authentication, and CRUD functionality became evident.

Step 2 — Database Design and Creation

Once the requirements were finalized, the MySQL database *cse_stock* was designed and created. This phase involved establishing a clean, normalized, and relational data structure.

2.1 Data Cleaning and Structure Standardization

The imported Excel sheets contained:

- Inconsistent column names
- Missing values
- Duplicate records
- Non-standard IDs (e.g., “F-001”, “F001”, “Chair1” mixed in one column)
- Unusable field types (TEXT instead of VARCHAR or INT)

These issues were corrected through:

- Renaming columns for clarity (e.g., “FurnName”, “quantity_ground”)
- Converting TEXT fields to appropriate types (VARCHAR for IDs, INT for quantities)
- Removing duplicates using SQL queries
- Adding default values and constraints

2.2 Adding Primary Keys and Relationships

Each table was assigned a proper Primary Key (PK) such as:

- *FurnID* for furniture tables
- *SL* for equipment tables
- *FacultyID* for faculty table

Foreign key relationships were created wherever logical, such as:

- Linking *FurnID* from quantity tables to furniture master table
- Maintaining referential integrity between assignment and faculty tables

2.3 Creating Useful SQL Views

To support analytics and dashboard reporting, two key SQL views were created:

- **vw_furniture_status** – shows discrepancy, serviceable count, ledger vs ground data
- **vw_equipment_status** – similar insights for equipment

These views simplified backend coding by performing calculations directly within SQL.

Step 3 — Backend Development Using Python

The backend logic was implemented in Python, focusing on database connectivity, CRUD operations, authentication, and data analytics.

3.1 Database Connectivity (mysql-connector-python)

A reusable database connection function was created to:

- Establish a secure connection to MySQL
- Handle incorrect credentials
- Manage exceptions during queries

3.2 CRUD Functionality

CRUD operations were implemented for both furniture and equipment tables:

- **Create** – Add new stock records
- **Read** – Fetch existing data for viewing or dashboards
- **Update** – Modify ledger quantity, ground quantity, or location
- **Delete** – Remove outdated or damaged items

All operations used parameterized queries to prevent SQL injection.

3.3 Authentication Module

A user authentication module was developed with:

- SHA-256 password hashing
- Login verification via MySQL
- Multiple roles (Admin, Lab Incharge, Viewer)
- Active/inactive user controls

This ensures only authorized users can update the inventory.

3.4 Reconciliation Logic (Ledger vs Ground)

Backend logic compared:

- *quantity_ledger*
- *quantity_ground*

And classified items as:

- **Matched**
- **Shortage**
- **Excess**

This was essential for audit reporting.

3.5 CSV Processing

Backend functions were created to:

- Upload CSV files containing ground stock
- Parse and validate data
- Compare values with ledger
- Generate discrepancy reports

This supports real departmental audits.

Step 4 — Frontend Development (HTML, CSS, JavaScript)

A fully modern, responsive frontend was built by the team using:

- HTML5
- CSS3 with custom UI styling
- Vanilla JavaScript
- Chart.js for visualizations

The frontend includes:

4.1 Dashboard

Displays:

- Total assets
- Furniture/equipment split
- Discrepancies
- Recent activity logs
- Bar and pie charts

4.2 Inventory Page

Includes:

- Search
- Filters
- Pagination
- Add/Edit/Delete modal
- Discrepancy badges (matched/short/excess)

4.3 Reconciliation Page

Supports:

- Uploading CSV
- Automated comparison
- Approve/Reject options
- Downloadable audit report

4.4 Faculty Assignment System

Allows mapping items to faculty members and tracking assignments.

4.5 Users & Settings

Supports role management, location/category updates, and password changes.

Step 5 — API Design for Integration

To integrate the backend with the frontend, API routes were designed for:

- Login
- Fetching inventory
- Adding/editing/deleting items
- Uploading CSV for reconciliation
- Viewing reports and dashboards
- Managing users and roles

Payload structures were finalized for:

- GET, POST, PUT, DELETE requests
- JSON-based data exchange

This step prepares the system for full integration.

Step 6 — Testing and Validation

Testing was performed at multiple levels:

6.1 Backend Testing

- SQL query validation
- CRUD operation testing
- Error handling for invalid inputs
- Login authentication testing

6.2 Frontend Testing

- UI responsiveness
- Chart rendering
- Modal functionality
- Navigation between pages

6.3 Integration Testing (Ongoing)

Tests ensure:

- API endpoints work as expected
- Frontend buttons submit real backend requests
- Discrepancy logic matches SQL views

6.4 Data Accuracy Checks

- Ledger vs ground matching
- Duplicate removal confirmation
- Record consistency across modules

Step 7 — Integration (Ongoing Work)

This final phase focuses on connecting frontend actions to backend logic:

- Replacing localStorage with MySQL-powered data
- Connecting login page with database authentication
- Linking Add/Edit/Delete buttons with backend CRUD APIs
- Using SQL views for dashboard charts
- Binding reconciliation results to real stock updates

Once completed, the system becomes a fully functional stock management platform.

Methodology (Design)

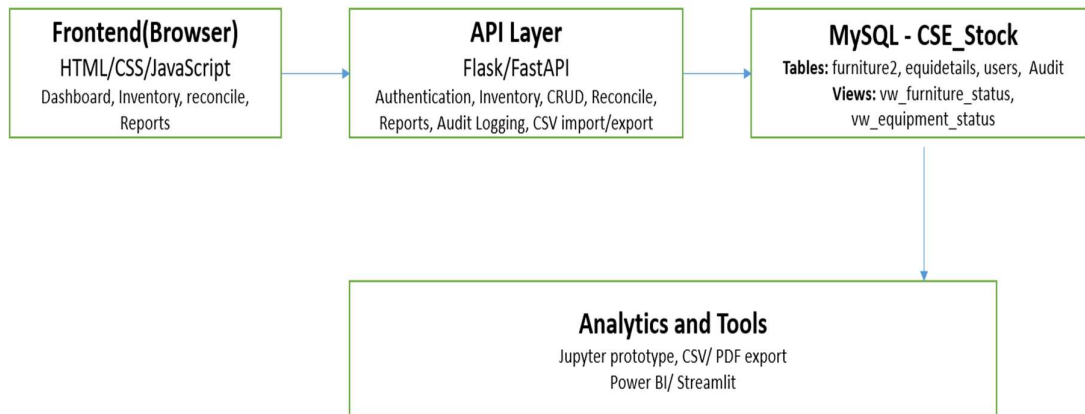


Fig No. : 1

Outcome

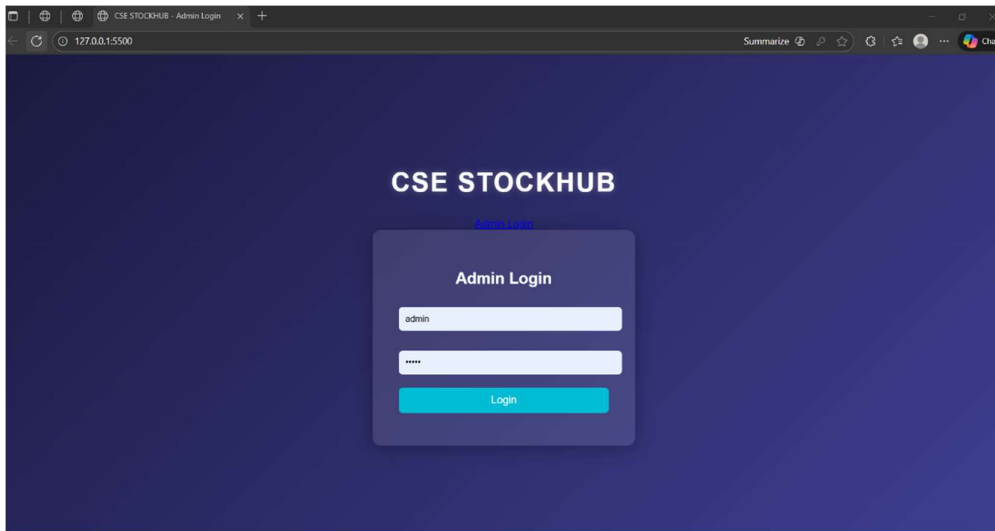


Fig 2: Login Page

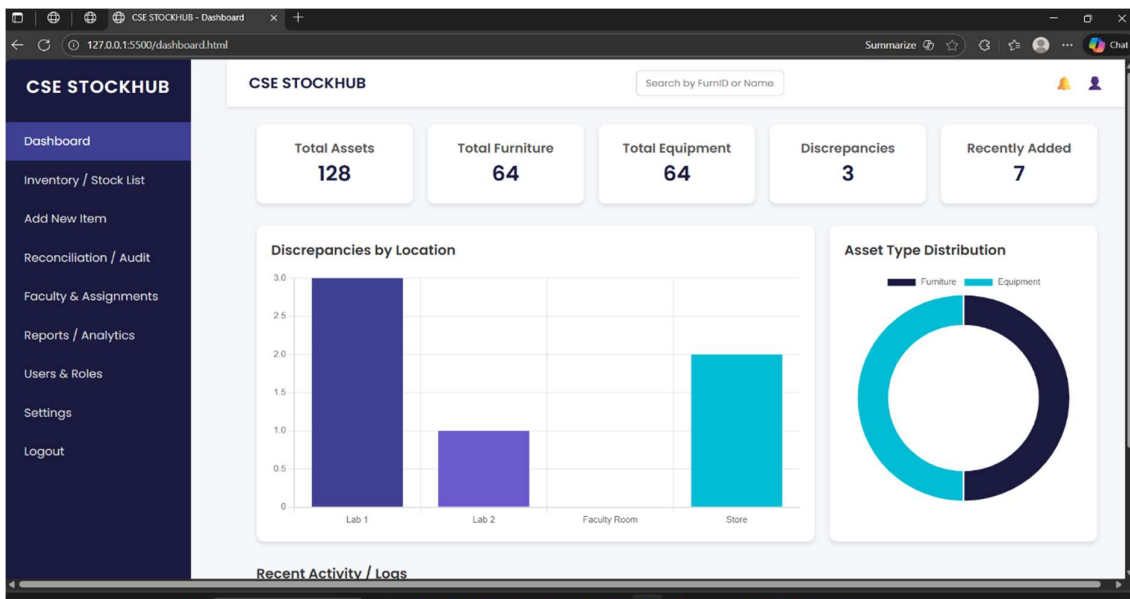


Fig 3: Dashboard

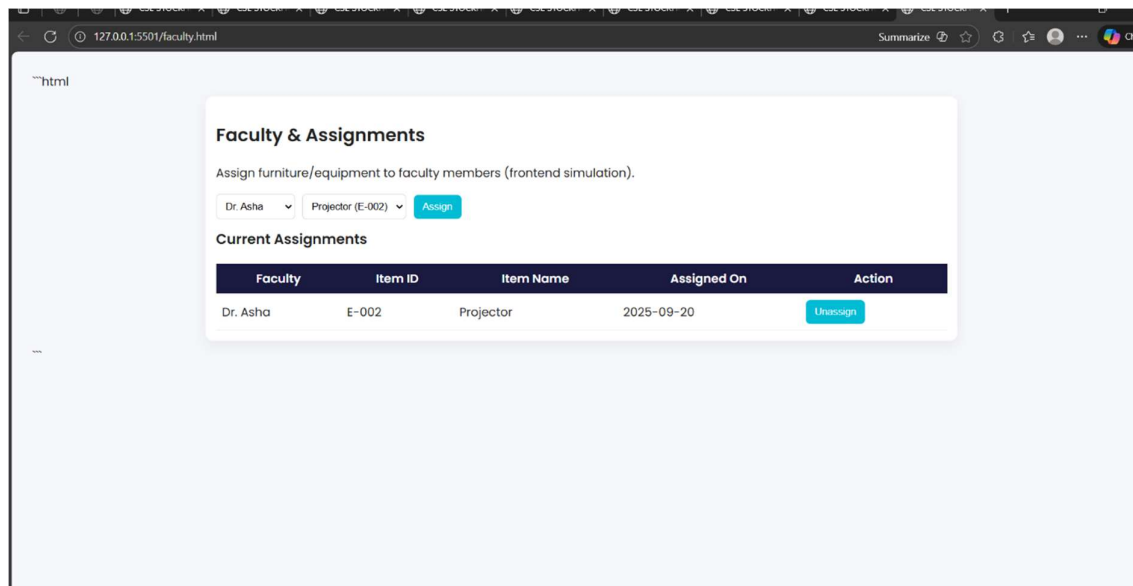


Fig 4: Faculty and Assignment

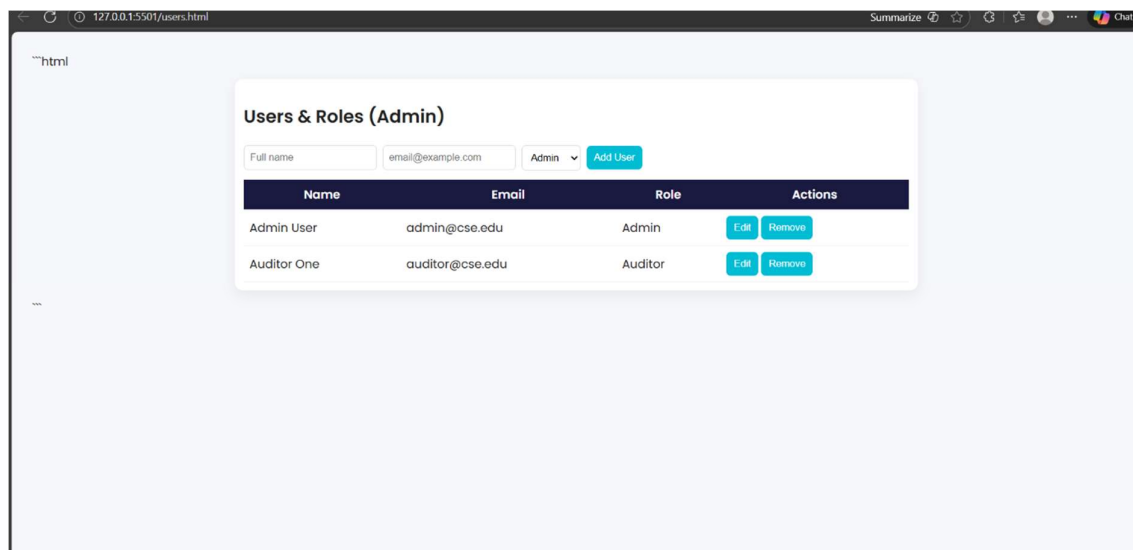


Fig 5: Users and Roles

The screenshot displays a web browser window with multiple tabs open, all titled 'CSE STOCKHUB'. The active tab is 'CSE STOCKHUB - Add / Edit Item', showing a form for adding or editing an item. The browser's address bar shows the URL '127.0.0.1:5501/add_edit.html'. The form itself is titled 'Add / Edit Item' and contains the following fields:

- Furniture / Equipment Name:** A text input field with a placeholder 'e.g., Projector'.
- FurnID:** A text input field with a placeholder 'e.g., E-010'.
- Ledger Quantity:** A numeric input field with the value '1'.
- Ground Quantity:** A numeric input field with the value '0'.
- Location:** A dropdown menu with 'Lab 1' selected.
- Type:** A dropdown menu with 'Furniture' selected.
- Remarks:** A text area with the placeholder 'Optional notes...'.

At the bottom right of the form, there are two buttons: 'Cancel' and 'Save Item'.

Fig 6: Data Entry Form

Remaining Work

Although the core frontend interface and backend database structure have been developed, several integration tasks remain to complete the CSE Stock Hub system. The frontend pages currently operate using dummy data and localStorage simulation; these need to be fully connected to the MySQL database through a dedicated backend API built using Flask or FastAPI. All inventory operations—including adding new items, editing existing entries, and deleting outdated records—must be linked to real database queries so that the UI reflects live departmental data. Additionally, the login page must be integrated with backend authentication logic to ensure that only authorized users with valid credentials and roles can access the system. Reconciliation features such as CSV-based stock verification should also be connected to the backend so that discrepancies can be processed, stored, and updated dynamically. Finally, comprehensive testing, validation of each module, and smooth integration of all pages are required to ensure that the entire system works reliably as a unified digital inventory management platform.

GANTT CHART

Activity	Time Frame				
	August	September	October	November	December
Literature Survey					
Problem Defination					
Design(UI/UX)					
Testing And Validation					
Documentation					



Proposed Activity



Activity Achieved



Ongoing Activity

REFERENCES

- [1] C. C. Wei, S. A. Ramiah, and N. F. Razali, “Inventory Management Systems (IMS),” *JATI Journal of Applied Technology and Innovation*, vol. 7, no. 3, pp. 15–22, 2023.
- [2] H. Wang, “Design of fixed assets management system in colleges and universities based on barcode and PFDI technology,” in *Journal of Physics: Conference Series*, vol. 1992, no. 3, pp. 1–7, 2021.
- [3] W. Kusuma, N. H. Adi, Afrina, and J. R. A. Jimenez, “Development of a web-based inventory information system to improve efficiency and effectiveness of inventory management,” *Journal of Computer-Based Instructional Media*, vol. 1, no. 2, pp. 82–98, Oct. 2023.
- [4] R. G. Raj and J. W. Wong, “Web-based asset management system for academic institutions,” *International Journal of Computer Applications*, vol. 180, no. 21, pp. 25–31, Feb. 2018.
- [5] S. A. Prabowo, L. Nugroho, and R. Hartanto, “Design and implementation of inventory management system using barcode technology,” *Procedia Computer Science*, vol. 135, pp. 377–384, 2018.